

CommitTrack

Complete Technical Documentation

What the app does · How every feature works · Full implementation reference

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Local-first · React · Vite · PWA · No backend

Table of contents (overview)

- Part I — Overview: purpose, architecture, stack, layout
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1. What is CommitTrack?

CommitTrack is a local-first Progressive Web App (PWA) for tracking personal financial commitments: recurring bills, EMIs, subscriptions, insurance premiums, money lent to or borrowed from people, savings goals, and month-by-month financial pressure.

All data is stored in the browser's localStorage on the user's device. There is no backend server, user authentication, or cloud sync in the current implementation. The app is built with React 19, Vite 8, React Router 7, Tailwind CSS 3, date-fns, and Recharts.

Primary audience: salaried individuals, freelancers, families, business owners, and students (student mode hides lending features). The app emphasizes privacy (data never leaves the device unless the user exports JSON/CSV), affordability awareness, and actionable insights rather than bank integration.

2. High-level architecture

- Entry: index.html !' main.jsx !' ErrorBoundary !' App.jsx
- App.jsx wraps CommitTrackProvider, BrowserRouter (with GitHub Pages basename), ThemeSync, routes
- Onboarding gate: settings.onboardingComplete must be true to see MainShell
- MainShell: Navbar + NotificationSync + lazy-loaded page routes inside Suspense
- State: CommitTrackContext holds commitments, lendings, settings, monthlySnapshots, goals
- Persistence: every mutation writes to localStorage immediately
- Intelligence: useCommitIntel hook aggregates 15+ engine modules for Home dashboard
- Engines: pure JavaScript functions (no React) for calculations and insights
- Utils: date handling, payments, migration, lending schedules, bill lifecycle

3. Technology stack

Layer	Technology	Purpose
UI	React 19	Components, hooks, context
Routing	React Router 7	SPA routes, lazy loading
Build	Vite 8	Dev server, production bundle
Styling	Tailwind CSS 3	Utility classes, dark mode class strategy
Dates	date-fns 4	Parsing, month arithmetic, formatting
Charts	Recharts 3	Analytics bar/line/pie charts
PWA	vite-plugin-pwa	Service worker, manifest, precache
Tests	Vitest 3	Unit tests for engines and utils
Deploy	gh-pages	Static hosting from dist/
PDF	pdfkit	This documentation generator

4. Repository layout

- src/pages/ — Full-screen route components (9 pages)
- src/components/ — Reusable UI including dashboard/, lending/ subfolders
- src/context/ — CommitTrackContext.jsx global state
- src/engines/ — Business logic calculators (21 modules)
- src/utils/ — Helpers, migration, repayment submodule
- src/constants/ — Categories, nav, modes, copy, symbols
- src/hooks/ — useCommitIntel, usePwaInstall, useProfileScope

- [src/services/notifications/](#) — Browser notification bridge
- [public/](#) — PWA icons, favicon
- [scripts/](#) — Build helpers, PDF generators

5. Application routes

5.1 Route table

Path	Component	Access
/	Home	Main shell, dashboard
/commitments	Commitments	Bill list, pay, edit
/add	Add	New bill form (not in nav)
/lending	Lending	Money page; ModeRoute blocks students
/analytics	Analytics	Charts; opened from Home month card
/tools	Tools	Redirects to /#dashboard-tools on Home
/profile	Profile	Settings, export, stats
/onboarding	Onboarding	First-run only shell
/lend/offer?d=...	LendingOfferReview	Outside main shell; share link

5.2 Home page (Dashboard)

- PageHeaderWithNotifications: bell opens NotificationPanel (portal, fixed position)
- InstallAppBanner: PWA install prompt
- HomeOverviewCard: gradient month card; tap navigates to /analytics
- Financial stability card: pressure score, health, yearly burden from useCommitIntel
- Goals preview: up to 3 goals with progress bars; Manage scrolls to #dashboard-tools
- Payoff priority, Insights, Forecast, Subscriptions cards when data exists
- Upcoming payments: top 3 pending active bills
- Overdue section: red-highlighted overdue bills
- DashboardTools: 4 calculator widgets at bottom (insurance, EMI, payoff, goals)
- ToolsDiscoveryToast: floating nudge on Home and Analytics; hides while scrolling

5.3 Commitments (Bills)

- Tabs/filters: All, Due now, Up next, Overdue, Paid history
- Sort: priority then due date
- Each row: category icon, name, Started/Ends/Due dates, amount, status badge
- Actions: tap row !' BillDetailModal; Pay opens payment modal; Edit !' CommitmentEditModal
- Add bill: navigates to /add
- Payment recording reduces remainingAmount; full pay on recurring spawns next cycle row

5.4 Add bill

- Category select drives insurance fields, interest rate, priority inference
- Start date (when bill began), End date (optional), Next payment due (computed from repeat)
- billDates.js: due date is next cycle "e today, not copy of start; end uses start MD + current year
- Repeat: none, monthly, bimonthly, quarterly, every4months, yearly
- Affordability card (mode-dependent): evaluateNewCommitmentAffordability
- Prior spend estimate for bills starting before current year

5.5 Lending (Money)

- Sections: Money you owe (borrowed) vs Money you lent
- Trust table per person from lendingTrustByPerson

- Manual add/edit modal with LendingFormFields + repayment schedule
- LendingRequestModal: borrower flow !' agreement !' share link
- LendingDetailModal: schedule, proofs, simulate UPI payment, HTML export
- Delete guarded by canDeleteLending when agreementLocked

5.6 Analytics

- Month at a glance: due, paid, left, free cash
- Affordability strip: income, burden, free after dues
- Biggest category pie data, highest recurring, 4-week due heatmap
- Lending repayment stats when lendings exist
- AnalyticsChartPanel: forecast, payments, categories + optional advanced charts

5.7 Profile

- ProfileAvatar: upload image or mode-based cartoon
- Collapsible Account settings: theme, income, user mode, active profile label
- Stats: streak, control score, payment count, recent payments
- Export JSON (all profiles) and CSV (scoped)
- Browser notification permission test
- No Analytics/Tools tiles (moved to Home)

5.8 Onboarding

- Select user mode (salaried, business, freelancer, family, student, power)
- Display name, monthly income, business type if applicable
- Sets onboardingComplete and persists settings

6. Data storage and schema (v8)

6.1 localStorage keys

Key	Type	Description
commitments	JSON array	All bill records
lendings	JSON array	All lending records
committrack_settings	JSON object	User preferences
committrack_monthly_snapshots	JSON array	Up to 48 monthly snapshots
committrack_goals	JSON array	Savings/debt goals
committrack_schema_version	number	Current: 8
committrack_tools_nudge_dismissed	flag	Calculator toast dismissed
committrack_pwa_install_dismissed	flag	Install banner dismissed
committrack_last_notif_digest	date	Daily notification digest

6.2 Commitment (bill) fields

- id, name, amount, remainingAmount, category, startDate, endDate, dueDate
- repeatType, priority (critical|medium|low), status, payments[{amount,date}]
- notes, profileId, annualInterestRate, trialEnd, priorSpend
- Insurance: insurancePolicyId, insuredPersonName, insuranceCompany, insuranceSumAssured, insuranceTermYears, insurancePremiumFrequency, insuranceMaturityBenefit
- createdAt, updatedAt timestamps

6.3 Lending fields

- id, personName, type (lent|borrowed), totalAmount, remainingAmount, dueDate
- payments with principalPortion, interestPortion, paymentType, onTime
- Agreement: agreementText, agreementLocked, offerId, borrowerSignName, lenderSignName, timestamps
- Financials from enrichLendingFinancials: interestRate, interestType, repaymentFrequency, repaymentSchedule[]
- proofs[], disputeStatus, relationshipTag, trustScoreSnapshot

6.4 Settings fields

- monthlyIncome, displayName, userMode, onboardingComplete
- activeProfileId (default|family|business labels only)
- colorScheme (light|dark|system), avatarSource, profileImageUrl
- readNotificationIds[], savedTowardGoals (legacy, migrated to goals)

6.5 Goal types

- reduce_open_debt: targetReduction amount
- income_ratio_cap: targetRatio (e.g. 0.45 = 45% of income to bills)
- save_amount: targetAmount + savedAmount progress

6.6 Monthly snapshot fields

- month (yyyy-MM), openRemainingSum, paidMonthSum, overdueSum, freeMoney
- pressureScore, commitmentRatio, monthlyBurden, totalCommitmentsCount, recordedAt
- Auto-appended once per calendar month from active profile commitments

7. CommitTrackContext API

7.1 Exposed data (profile-scoped unless noted)

- commitments, sortedCommitments — filtered by activeProfileId
- lendings, goals — filtered
- allCommitments, allLendings, allGoals — unfiltered
- settings, monthlySnapshots, todayStr, activeProfileId

7.2 Status helpers

- getEffectiveStatus(commitment) — paid|upnext|pending|overdue based on dates and remaining
- getEffectiveLendingStatus(lending) — pending|overdue|complete

7.3 Mutations

- addCommitment, updateCommitment, deleteCommitment
- addCommitmentPayment — applies payment; advanceRecurringCommitment on full pay
- addLending, updateLending, deleteLending (with agreement lock rules)
- addLendingPayment — structured principal/interest split
- updateSettings, addGoal, updateGoal, deleteGoal, logSavingsToGoal
- markNotificationRead, markAllNotificationsRead

8. Bill lifecycle and status engine

8.1 `getEffectiveStatus (commitmentStatus.js)`

- If `endDate` passed !' paid
- If `remainingAmount` ≤ 0 and non-recurring !' paid
- If `due month` $>$ current month !' upnext
- If `dueDate` $<$ today and `remaining` > 0 !' overdue
- Else !' pending

8.2 Recurring advancement (`commitmentRecurring.js`)

- `advanceRecurringCommitment`: when remaining hits 0 on repeating bill
- Creates paid row + new pending row with `dueDate` advanced by repeat interval
- `repeatType` none: single paid row only

8.3 `isBillDueInMonth (repeatTypes.js)`

- Determines if obligation exists in calendar month for analytics/forecast
- Yearly: matches month-day of due/anchor
- Other repeats: month offset divisible by interval

8.4 `billDates.js`

- `defaultEndDateFromStart`: same MM-DD as start, current year ("e start)
- `defaultDueDateFromStart`: next due on/after today from anchor + repeat
- `applyBillStartDateChange` / `applyBillRepeatChange` for Add/Edit forms

8.5 Payments (`commitmentPayments.js`)

- `applyPaymentToCommitment`: adds payment, reduces remaining, may update status

9. Calculation engines (complete reference)

9.1 `affordability.js`

- `evaluateAffordability(income, burden)` !' tier, label, `committedPercent`, `freeMoney`
- `evaluateNewCommitmentAffordability` — simulates adding draft bill
- `affordabilityBadgeClass(tier)` — Tailwind classes for UI badge

9.2 `burden.js`

- `monthlyBurdenForCommitment` — amount per month for one bill
- `totalMonthlyBurden` — sum across open commitments
- `monthlyBurdenForDraft` — for affordability preview

9.3 `pressureScore.js`

- `computeCanonicalPressureScore` — 0-100 from burden, overdue, snapshots
- `freeMoneyAfterBurden` — income minus monthly burden, open remaining
- `pressureScoreLabel`, `pressureScoreBadgeClass`

9.4 `pressureAdvanced.js`

- commitmentToIncomeRatio, monthlyPressureScore, pressureSeverity

- yearlyBurdenEstimate — annualized burden projection

9.5 financialHealth.js

- computeFinancialHealthScore — combines burden, overdue count, open debt
- healthLevelBadgeClass

9.6 intelligence.js

- generateCommitmentInsights — core dashboard insight cards with tone (critical/warning/positive)

9.7 insightsExtended.js

- overlappingDueDatesInsight, forecastCrunchInsight
- subscriptionYearlyCostInsight, emiBurdenPercentInsight
- mergeExtendedInsights — dedupe and cap count

9.8 forecast.js & forecastSeries.js

- forecastInsights — short text forecasts for dashboard
- buildCashflowForecastSeries — 12-month due vs free cash bars
- scheduledGrossInMonth, amountDueInMonth

9.9 analyticsSeries.js

- snapshotsToPressureTrend, debtReductionFromSnapshots
- recurringGrowthSeries, freeCashflowTrend
- buildDueHeatmap — 4-week buckets
- lendingPrincipalInterestTotals, lendingCompletionStats
- categoryOpenTrend (exported, limited UI use)

9.10 payoffOptimizer.js

- debtsFromCommitments, snowballOrder, avalancheOrder
- estimatePayoffTimeline, comparePayoffStrategies
- Recommendation with reason for DashboardTools payoff modal

9.11 payoffPriority.js

- effectiveAnnualRate by category default
- payoffPriorityScore, rankPayoffOrder, topPayoffRecommendation

9.12 prepayment.js

- simulatePrepayment — months saved, interest saved
- computeEmiFromPrincipal — standard EMI formula

9.13 insuranceCalculator.js

- simulateInsurancePolicy — premiums, maturity, IRR-style metrics
- analyzeInsuranceWorth — inflation-adjusted verdict
- insuranceParamsFromBill

9.14 goalsProgress.js

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- computeGoalProgress — 0-1 fraction by goal type
- goalTypeLabel

9.15 lendingTrust.js

- lendingTrustByPerson — aggregate on-time/delayed per person
- trustScoreForLendingEntry, trustBadgeClass, trustSummaryLine

9.16 lendingAgreement.js

- buildAgreementText — plain-language loan terms
- encodeOfferPayload / decodeOfferPayload — base64 share link
- buildOfferShareUrl — /lend/offer?d=
- borrowerTrustSnapshot, canDeleteLending, trustScoreLabel

9.17 reminders.js & notifications.js

- buildSubscriptionEndReminders, buildCommitmentReminders, buildLendingReminders
- buildNotificationFeed — merges reminders + insights for bell
- unreadCount

9.18 snapshots.js & subscriptionLeak.js

- buildMonthlySnapshot — captures month KPIs
- subscriptionLeakReport — low-priority recurring spend signals

10. Lending flows (step-by-step)

10.1 Manual entry

- User opens Money !' Add !' fills LendingFormFields
- buildLendingRecord + enrichLendingFinancials generates repaymentSchedule
- addLending persists normalized row

10.2 Borrower request flow

- LendingRequestModal step 1: names, amount, rate, due, collateral, purpose
- Step 2: agreement text, borrower signature, accept checkbox
- Creates type=borrowed with agreementLocked, offerId
- buildOfferShareUrl copies link for lender

10.3 Lender offer acceptance

- LendingOfferReview at /lend/offer?d=payload
- decodeOfferPayload shows terms + borrower trust score
- Lender signs !' addLending type=lent with mirrored agreement fields

10.4 Repayment submodule (utils/repayment/)

- schedule.js — generate installment rows by frequency
- calculations.js — EMI, simple/compound interest
- payments.js — applyPaymentToLending with allocation
- index.js re-exports

11. UI components inventory

11.1 Core

- Card — surface container; ui-card class when enhanced theme
- Modal — titled dialog with footer slot
- Navbar — 4-tab mobile grid, desktop top bar
- PageHeaderWithNotifications — title + bell
- NotificationPanel — portal overlay reminder list
- ThemeSync — applies dark class from settings.colorScheme
- ModeRoute — redirects if path not allowed for user mode
- ErrorBoundary — catches render errors

11.2 Bills

- BillDetailModal — spend summary, payment history, insurance hint
- CommitmentEditModal — full edit with billDates helpers
- CategoryChip, PriorityBadge, InsuranceFields

11.3 Dashboard

- HomeOverviewCard, DashboardTools, ToolsDiscoveryPrompt
- ToolWidget, InsuranceCalculatorModal
- RoleDashboardPanel (defined, optional future use)

11.4 Lending

- LendingFormFields, LendingRequestModal, LendingDetailModal
- LendingDetailDashboard — stats, timeline, schedule table

11.5 Other

- AnalyticsChartPanel — Recharts switcher
- ProfileAvatar, CollapsibleSection, InstallAppBanner

12. Hooks and services

12.1 useCommitIntel

- Aggregates: stability, health, insights, forecast, subscriptionLeak
- payoffRec, notifications, notificationUnread, yearlyBurden, burdenRatio
- Used by Home, NotificationPanel, NotificationSync

12.2 usePwaInstall

- Captures beforeinstallprompt, exposes install(), dismiss, platform hints

12.3 Notification services

- browserNotifications.js — permission, showNotification
- scheduler.js — daily digest max 3 items
- reminderBridge.js — connects intel reminders to browser API
- NotificationSync — runs scheduler on interval when app open

13. Categories, priority, and user modes

13.1 Bill categories

- EMI, Credit Card, Subscription, Insurance, SIP, Rent, Loan, Utility, Other
- Insurance shows InsuranceFields; EMI/Loan show interest %
- Subscription supports trialEnd and end-date cancel reminders

13.2 Priority

- critical, medium, low — inferred from category except Other (user picks)

13.3 User modes

Mode	Nav tabs	Lending	Affordability on Add
salaried	4	Yes	Yes
business	4	Yes	No
freelancer	4	Yes	Yes
family	4	Yes	Yes
student	3 (no Money)	No	Yes
power	4	Yes	Yes

14. PWA and deployment

14.1 PWA

- vite-plugin-pwa: autoUpdate service worker
- Precaches JS, CSS, HTML, icons
- manifest: standalone, portrait, theme color
- InstallAppBanner + usePwaInstall

14.2 Deployment

- npm run build !' dist/
- gh-pages deploy; basePath /PROJECTapp/ for GitHub Pages
- copy-404.mjs duplicates index.html for SPA routing

14.3 UI theme (revertible)

- src/constants/uiTheme.js — UI_THEME_ID: enhanced | legacy
- enhanced loads ui-enhanced.css: Plus Jakarta Sans, Outfit, gradients
- Set legacy to revert fonts/colors without code rollback

15. Automated tests

15.1 Test files (15 tests)

File	What it tests
affordability.test.js	Income ratio tiers and free money
payoffOptimizer.test.js	Snowball vs avalanche ordering
commitmentStatus.test.js	Overdue and paid status
commitmentRecurring.test.js	Monthly cycle advancement
repayment.test.js	EMI schedule and payments
billDates.test.js	Due/end date defaults from start

16. Implementation status, gaps, and roadmap

16.1 Fully implemented

- Full bill CRUD with recurring cycles and payments
- Lending with schedules, trust, agreements, share links

- Home dashboard intelligence and calculator tools
- Analytics with 6 chart views
- Profile export, theme, avatars, onboarding
- PWA install shell and in-app notifications

16.2 Partial / basic

- Profiles: filter labels only, no create/rename UI
- Snapshots: global per month, not per profile
- Notifications: no push when app closed; daily digest only
- Goals: no edit modal, only add/delete/log savings
- No JSON import (export only)
- Mutual cancel fields exist but limited UI

16.3 Not implemented

- Backend, authentication, cloud sync, real UPI
- Bank feed / SMS parsing
- E2E tests, CI pipeline in repo
- CommitTrack Plus (placeholder)
- Legal e-sign compliance beyond local typed names

16.4 Suggested next steps

- JSON import paired with export
- Per-profile monthly snapshots
- Background due reminders
- Include lending in 12-month cashflow forecast
- Goal edit UI and onboarding re-entry from Profile

17. Utilities module reference

17.1 dates.js

- todayYmd() — local calendar date YYYY-MM-DD
- compareYmd(a,b) — string compare for dates

17.2 migrateStorage.js

- CURRENT_SCHEMA_VERSION = 8
- normalizeCommitment, normalizeLending, normalizeGoal
- loadInitialAppState — reads all keys, migrates legacy savedTowardGoals
- saveMonthlySnapshotsToStorage, saveGoalsToStorage

17.3 commitmentStatus.js

- getEffectiveStatus(c, todayStr)
- normalizeCommitmentStatusForSave — persists computed status

17.4 commitmentPayments.js & commitmentSpendSummary.js

- applyPaymentToCommitment, totalPaidOnPayments
- summarizeCommitmentSpend — spent, future, total for BillDetailModal

17.5 billLifecycle.js & billDisplayName.js

- estimatePriorSpend, BILL_STATUS_UI, isActiveBill
- display name helpers for insurance bills

17.6 lendingRecord.js & lendingFinancials.js

- buildLendingRecord — canonical new lending object
- enrichLendingFinancials — schedule, principal, interest breakdown

17.7 lendingStatus.js & lendingPayments.js & lendingTimeline.js

- getEffectiveLendingStatus, applyPaymentToLending
- buildLendingTimeline events for detail dashboard

17.8 monthPaymentSummary.js

- computeCurrentMonthSummary — due/paid/left/free for Analytics month card

17.9 profileScope.js & profileStats.js

- filterByProfile(list, profileId)
- computePaymentMonthStreak, computeControlScore, outstandingLent

17.10 theme.js & applyUiTheme.js & basePath.js

- bootstrapThemeFromStorage — applies dark class before paint
- routerBasename for GitHub Pages /PROJECTapp/

17.11 agreementExport.js

- exportAgreementHtml — downloadable HTML summary of lending agreement

18.1 Key constant files

- categories.js — CATEGORIES, categoryShowsInsuranceFields, categoryShowsInterestRate
- repeatTypes.js — REPEAT_OPTIONS, isBillDueInMonth, normalizeRepeatType
- priority.js — inferPriorityFromCategory, OTHER_PRIORITY_OPTIONS
- insurance.js — buildInsuranceBillName, emptyInsuranceFields, repeatTypeToPremiumFrequency
- nav.js — NAV_ITEMS (Home, Bills, Money, Profile)
- userModes.js — USER_MODES, navItemsForMode
- plainLanguage.js — CHART_VIEWS, TOOLS_PLAIN, legal disclaimer
- symbols.js — formatInr, INR, TOOL_ICONS, STATUS_ICONS
- copy.js — user-facing strings (addBill, recordPayment, etc.)
- profileAvatars.js — cartoon avatars per user mode

19. Complete source file inventory

Every .jsx/.js file under src/ is listed below by folder. Files marked (test) are Vitest specs.

19.1 Pages (9)

- Home.jsx, Commitments.jsx, Add.jsx, Lending.jsx, LendingOfferReview.jsx
- Analytics.jsx, Tools.jsx, Profile.jsx, Onboarding.jsx

19.2 Components (28)

- AnalyticsChartPanel, BillDetailModal, Card, CategoryChip, CollapsibleSection
- CommitmentEditModal, ErrorBoundary, InstallAppBanner, InsuranceCalculatorModal
- InsuranceFields, LendingDetailModal, Modal, ModeRoute, Navbar
- NotificationPanel, NotificationSync, PageHeaderWithNotifications
- PriorityBadge, ProfileAvatar, ThemeSync, ToolWidget
- dashboard/: DashboardTools, HomeOverviewCard, RoleDashboardPanel, ToolsDiscoveryPrompt
- lending/: LendingDetailDashboard, LendingFormFields, LendingRequestModal

19.3 Engines (21 + 2 tests)

- affordability, analyticsSeries, burden, financialHealth, forecast, forecastSeries
- goalsProgress, insightsExtended, insuranceCalculator, intelligence
- lendingAgreement, lendingTrust, notifications, payoffOptimizer, payoffPriority
- prepayment, pressureAdvanced, pressureScore, reminders, snapshots, subscriptionLeak

19.4 Utils (24 + 4 tests)

- agreementExport, applyUiTheme, basePath, billDates, billDisplayName, billLifecycle
- commitmentPayments, commitmentRecurring, commitmentSpendSummary, commitmentStatus
- dates, formatBurdenPercent, lendingFinancials, lendingPayments, lendingRecord
- lendingStatus, lendingTimeline, migrateStorage, monthPaymentSummary
- profileScope, profileStats, theme
- repayment/: calculations, payments, schedule, index

20. Data flow diagrams (narrative)

20.1 Adding a bill

- 1. User fills Add.jsx form !' validate()
- 2. handleSubmit builds draft object with priorSpend estimate
- 3. addCommitment !' normalizeCommitment !' normalizeCommitmentStatusForSave
- 4. localStorage.setItem('commitments')
- 5. React state updates !' Commitments list re-renders
- 6. Next month: useEffect may append monthlySnapshots entry

20.2 Recording a payment

- 1. User enters amount/date on Commitments
- 2. addCommitmentPayment(id, {amount, date})
- 3. applyPaymentToCommitment reduces remaining
- 4. If remaining=0 and recurring: advanceRecurringCommitment returns [paidRow, nextCycle]
- 5. flatMap replaces one row with one or two rows in storage

20.3 Intelligence pipeline

- 1. useCommitIntel reads commitments, settings, snapshots
- 2. Runs engines: pressure, health, insights, forecast, notifications
- 3. Home renders cards from intel.* properties
- 4. NotificationSync may fire browser notifications from reminders

21. Key formulas and algorithms

21.1 EMI (prepayment.js)

$EMI = P \times r \times (1+r)^n / ((1+r)^n - 1)$ where $r = \text{annualRate}/12/100$, $n = \text{months}$. Prepayment simulation adds extra monthly principal to reduce months and interest.

21.2 Pressure score

Canonical score 0–100 combines: ratio of monthly burden to income, overdue count weight, trend from last two snapshots if available. Label bands: Comfortable, Watch, Stressed, Critical.

21.3 Trust score (lending)

Per person: weighted on-time vs delayed repayments across historical lending entries. Displayed 0–100 on offer review and lending cards.

21.4 Goal progress

reduce_open_debt: progress vs baseline open remaining reduction target

income_ratio_cap: progress toward keeping burden ratio below target

save_amount: savedAmount / targetAmount

